CHRIS KRANER

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Physics-trained analyst and educator specializing in automation, quantitative modeling, and instructional design. Builds streamlined workflows, data-driven evaluation tools, and hands-on learning resources that clarify complex information and strengthen decision-making. Focus on creating adaptable digital solutions that improve operational efficiency and program outcomes.

CORE COMPETENCIES

Data & Analytics	Technology & Process Improvement	Communication & Training
• Quantitative Modeling	• Workflow Automation	• Instructional Design
• Program Evaluation	• Process Optimization	• Stakeholder Engagement
• Data Visualization	• Requirements Analysis	• Workshop Development
• Data Pipelines & Wrangling	• Systems Integration	• User Adoption Support
• Analytical Reporting	• Digital Solution Development	• Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE & PROJECTS

Personal Project

- Built a frequency-domain model of outdoor HVAC sound propagation using wave-based physics for diffraction, interference, and ground reflection.
- Produced a structured technical paper combining physics reasoning with AI-assisted verification, ready for translation into modular Python code.
- Artifact available on GitHub

Educational Assessment Research

- Conducted statistical analysis using R to evaluate pre-service teacher perceptions of assessment practices and compare findings with international educational models.
- Identified meaningful differences between U.S. and international assessment frameworks and developed alternative models to better represent domestic educational perspectives.
- Prepared technical reports and research findings to support academic decision-making and future initiatives.
- Artifact available on GitHub

Illinois Pathways Initiative

- Analyzed educational pathways across high schools, community colleges, and universities to identify efficient degree-completion routes for high-demand career fields.
- Coordinated with multiple institutions to validate transfer credits, curriculum requirements, and pathway alignment.
- Developed student- and parent-facing process documentation and guides using Adobe Creative Suite.
- Evaluated data management and integration challenges, identifying operational limitations and opportunities for process improvement.

Technology Innovation Project – Google Glass Integration

- Supported the setup and use of a technology-assisted app that delivers real-time speech pacing feedback for students.
- Leveraged Android Debug Bridge (ADB) tools to adapt existing applications for educational use cases.
- Collaborated with faculty and educators to support technology adoption through demonstrations, training sessions, and instructional job aids.
- Artifact available on GitHub

EDUCATION

- Northern Illinois University – Graduate Certificate in Quantitative Methods
- Purdue University – Bachelor of Science in Physics

TECHNICAL SKILLS

Programming & Development	Python | JavaScript | HTML/CSS | Flask | Component-Based Web Frameworks
Data Analysis & Presentation	R | Factor Analysis | UX/UI Testing | Survey Methods | Qualitative Methods | Outcome Measurement | Validation | Data Visualization | Dashboard Creation
Content & Documentation	Adobe Creative Suite | Technical Documentation | Process Guides | Digital Content Editing | Technology Onboarding
Instructional Design & Applied Learning	K12 | College | Educator | Community

TRAINING, FACILITATION & PROGRAM SUPPORT

- Designed and delivered workshops, classes, and outreach programs for diverse audiences.
- Developed lesson plans, instructional content, and learning materials aligned with organizational objectives and learner needs.
- Utilized surveys, interviews, and usability testing methodologies to assess effectiveness and measure program impact.
- Created automation and visualization tools using Python, JavaScript, HTML/CSS, and Flask to improve operational efficiency and support educational initiatives.